Table 1: Curriculum Model Average (CMA) exhibits advantages over standard LR decay schedule pretraining, much better than the widely used *Cosine+Uniform* setting. Our proposed methods are highlighted in gray. **WA**: Weight Averaging technique (Section 3). **Order**: Data ordering. **LRS**: Learning Rate Schedule (WSD: Warmup-Stable-Decay, Cos: Cosine, Const: Constant). **Core**: Average score on the first four, high signal-to-noise tasks according to prior work (Heineman et al., 2025) (MMLU, ARC-c, ARC-e, CSQA). Both the Core and Avg. scores are annotated with a subscript indicating the performance change relative to the baseline (*WSD + Uniform*). Performance changes are color-coded: **bold green** (≥ 0.5 improvement), light green (> 0 improvement), and red (decrease).

WA	Order	LRS	MMLU	ARC-c	ARC-e	CSQA	Core	OBQA	PIQA	SIQA	Wino.	Avg.
✗	Uniform	Cos	30.49	38.13	59.47	49.14	$44.31_{-1.90}$	42.20	71.87	45.19	56.51	$49.13_{-1.43}$
✗	Ascend	Cos	30.80	39.80	59.12	51.27	$45.25_{-0.96}$	42.60	71.55	45.65	57.06	$49.73_{-0.83}$
✗	Uniform	WSD	30.77	42.14	61.05	50.86	46.21	45.20	72.42	45.75	56.27	50.56
✗	Ascend	WSD	31.58	38.80	61.05	50.37	$45.45_{-0.76}$	45.80	71.82	46.01	57.30	$50.34_{-0.22}$
WMA	Uniform	Const	30.87	37.12	58.95	53.24	$45.04_{-1.17}$	43.40	71.76	46.26	57.38	$49.87_{-0.69}$
SMA	Uniform	Const	31.22	36.12	59.82	53.97	$45.28_{-0.93}$	43.40	71.98	46.42	57.85	$50.10_{-0.46}$
EMA	Uniform	Const	31.39	36.45	59.82	53.48	$45.29_{-0.92}$	42.40	72.14	46.32	57.54	$49.94_{-0.62}$
WMA	Ascend	Const	31.67	39.80	61.40	53.07	$46.49_{+0.28}$	45.00	71.93	45.45	57.14	$50.68_{+0.12}$
SMA	Ascend	Const	32.28	40.80	62.11	52.91	$47.02_{\mathbf{+0.81}}$	44.80	71.60	45.80	57.22	$50.94_{+0.38}$
EMA	Ascend	Const	32.17	40.80	61.75	53.07	$46.95_{\mathbf{+0.74}}$	44.80	71.55	45.85	57.62	$50.95_{+0.39}$

riculum. These results support our hypothesis: when the LR decays, folding offers a slight benefit over simple sorting by processing some high-quality data earlier, but under a constant LR, where this is not a concern, a simple end-to-end curriculum remains superior. Refer to Section B.1 for a more detailed discussion on data folding experiments.

5 Unlocking Data Curricula Potential via Moderate LR Decay and Model Averaging

To resolve the interaction between data curricula and learning rate (LR) schedules, we first utilize a more moderate LR decay in place of a standard aggressive decay, which we find mitigates the issue. We then turn to a more principled approach: *model averaging* (Izmailov et al., 2018; Li et al., 2025c; Tian et al., 2025). We investigate replacing LR decay entirely with model averaging, which allows high-quality data to be processed with a constant learning rate. While model averaging alone may not match the performance of LR decay with uniform data, we find that combining a data curriculum with model averaging produces comparable or even superior results to a standard decaying LR schedule, particularly in a mid-training setting. This reveals a synergistic relationship between data curricula and model averaging. Furthermore, we find that a combination of model averaging and moderate LR decay can yield even stronger and stable results for curriculum-based pretraining. Our results highlight a previously unexplored regime for improving LLM pretraining and reveal the potential of co-designing LR schedules, data curricula, and model averaging strategies.

5.1 Mitigating Negative Interaction with Moderate Learning Rate Decay

A straightforward way to mitigate the negative impact of LR decay on data curricula is to use a moderate LR decay instead of a standard aggressive one. As shown in Figure 2, for uniform data, the validation loss decreases with more aggressive LR decay, like increasing decay steps and lower ending LRs in our experimental setting. In comparison, the benefits of data curricula increase with a more moderate LR decay, like fewer decay steps or higher ending LRs. The different preferences of LR decay suggest that the optimal ending LR or number of decay steps can differ for curriculum-based training from uniform data training, also indicated by the validation loss curves in Figure 2. Since the optimal ending LR is typically close to zero for uniform data ordering, but more decay steps are not always better (Li et al., 2025b; 2024; Hu et al., 2024), it is more convenient to ablate on the ending LRs to see whether the optimal LR schedule changes for curricula.

To investigate this, we run experiments on ending LRs in a fine-grained manner and report the training results for both uniform data ordering and a data curriculum. As shown in Figure 5(a), we find that the data curriculum (Ascend+WSD) may only achieve a marginal improvement or even fail to match the performance of uniform ordering (Uniform+WSD) when the ending LR is close to zero, like at the scale of 10^{-5}. However, as the ending LR increases, the performance of curriculum training improves, but may degrade when the ending LR approaches the peak LR. Note that the performance of the data curriculum can decline while its relative benefit over uniform ordering can still increase as the uniform training performance degrades more sharply when ending LR increases. The optimal ending LR of the data curriculum is around 1×10^{-3}, much higher than that for uniform

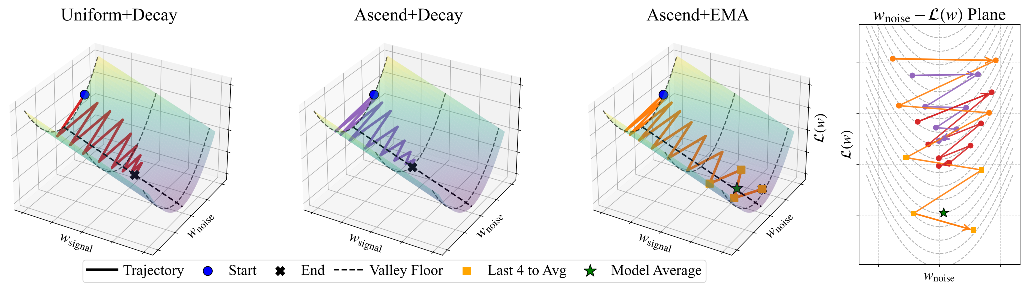

Figure 4: Visualization of our intuition about the interplay between data ordering and LR schedules. We assume the gradient update can be decomposed as a signal direction and a noise direction. High-quality data can offer a less noisy direction and a more stable signal direction, while low-quality data can induce a more noisy update. *Uniform+Decay*, *Ascend+Decay* and *Ascend+EMA* represent different training strategies. *Ascend+EMA* can make the best use of the high-quality data in the curriculum. The right-hand figure shows the projection of the trajectories of the last 8 steps for these cases onto the w_{noise}-$\mathcal{L}(w)$ plane.

data ordering, and the tuned data curriculum training outperforms the optimal uniform data training results. These experiments validate that using a more moderate LR decay—for instance, adjusting the ending LR to approximately $1/3$ of the peak LR in our setting—can effectively mitigate the negative interaction and unlock the benefits of a data curriculum.

5.2 Model Average can Help Data Curriculum

CMA: Replacing Learning Rate Decay with Model Averaging. Adjusting the ending LR serves as a trade-off between the benefits of a data curriculum and LR decay. To fully utilize the benefits of data curricula, we propose to decouple the data schedule from the side effects of LR annealing by replacing LR decay entirely with model averaging. In this approach, we replace the decaying LR schedule with a constant LR and apply model averaging to the final checkpoints of the training process. We call this strategy **C**urriculum **M**odel **A**veraging (CMA), detailed in Algorithm 1. In our default setting, we use Exponential Moving Average (EMA) with $\alpha = 0.2$ over the last six checkpoints. The types of weight averaging considered include Simple Moving Average (SMA), EMA, and Weighted Moving Average (WMA), as introduced in Section 3. For comparison, we use standard LR pretraining schedules, including cosine and WSD (introduced in Section 3). The strongest baseline for our evaluation, denoted *WSD + Uniform*, is the best-performing combination of a standard LR schedule and data order. Downstream task performances are reported in Table 1, and practical implementation details are reported in Section A.2.

The Synergy of Data Curriculum and Model Averaging. The results in Table 1 lead to several key observations. First, the combination of a data curriculum and model averaging (e.g., *EMA + Ascend*) outperforms models trained with a standard LR decay schedule, including *WSD + Uniform* and *WSD + Ascend*. It also consistently outperforms model averaging applied to a uniform data order (e.g., *EMA + Uniform*). Second, this synergy is crucial, as other combinations yield limited improvements. For instance, model averaging with a uniform data order (*EMA + Uniform*) under a constant learning rate does not fully match a standard WSD schedule. Furthermore, combining a standard LR decay with a data curriculum (*WSD + Ascend*) provides only marginal gains and can even degrade performance, confirming the negative interaction we identified between schedules. These results highlight the necessity of combining both a data curriculum and weight averaging, a strategy largely overlooked by prior work that has focused on either weight averaging (Tian et al., 2025; Yang et al., 2024b) or curriculum design (Dai et al., 2025) in isolation. Third, aligning the checkpoint weights with the data schedule is beneficial: EMA and SMA, which assign non-decreasing weights to later (and higher-quality) checkpoints, outperform WMA under a data curriculum. The differences between these moving average strategies are detailed in Section 3.

Motivation: Synergy from Decoupling Schedules. We intuitively interpret the synergy between a data curriculum and model averaging from a loss landscape perspective, emphasizing the interplay between two key factors: the learning rate and data quality. We present a visualization of this concept in Figure 4. The learning rate controls the size of the update steps, while data quality influences the signal-to-noise ratio of the gradients. *Uniform+Decay* progresses with a relatively consistent level of noise compared to the signal from the training data, and the final decay reduces the noise but also limits the update rate along the signal direction; *Ascend+Decay* starts with high noise, but the step-size decays too fast, so it does not utilize the good signal from high-quality data to move faster near the end; When it comes to the model averaging strategy, the training over

Table 2: The benefit of CMA becomes more prominent in the mid-training setting. Our proposed methods are highlighted in gray. **WA**: Weight Averaging technique (Section 3). **Order**: Data ordering in two phases (U: Uniform, A: Ascend). A-T (All-Together) sorts data samples in both phases as a whole. **LRS**: Learning Rate Schedule (WSD: Warmup-Stable-Decay schedule, Const: Constant LR). **Core**: Average score on the first four, high signal-to-noise tasks (MMLU, ARC-c, ARC-e, CSQA). Both the Core and Avg. scores are annotated with a subscript indicating the performance change relative to the baseline (*WSD + U,U*). Performance changes are color-coded: **bold green** ($\geq$ 0.5 improvement), light green ($>$ 0 improvement), and red (decrease).

WA	Order	LRS	MMLU	ARC-c	ARC-e	CSQA	Core	OBQA	PIQA	SIQA	Wino.	Avg.
✗	U,U	WSD	29.23	33.78	53.86	49.55	41.61	40.40	71.87	44.78	56.43	47.49
✗	U,A	WSD	29.44	34.45	52.63	50.12	$41.66_{+0.05}$	41.00	71.76	44.42	56.75	$47.57_{+0.08}$
✗	A,A	WSD	30.22	33.11	56.84	47.34	$41.88_{+0.27}$	39.40	71.55	44.78	56.67	$47.49_{0.00}$
✗	A-T	WSD	29.93	37.12	54.39	49.47	$42.73_{\mathbf{+1.12}}$	39.00	72.20	45.14	56.83	$48.01_{\mathbf{+0.52}}$
EMA	U,U	Const	29.84	32.78	52.28	51.52	$41.60_{-0.01}$	42.00	71.60	44.68	56.99	$47.71_{+0.22}$
EMA	U,A	Const	29.75	35.12	51.75	48.57	$41.30_{-0.31}$	42.20	70.51	44.83	56.91	$47.45_{-0.04}$
EMA	A,A	Const	30.31	36.45	57.54	50.12	$43.61_{\mathbf{+2.00}}$	41.40	72.14	45.09	56.43	$48.69_{\mathbf{+1.20}}$
EMA	A-T	Const	30.81	36.29	57.89	50.29	$43.82_{\mathbf{+2.21}}$	44.50	70.62	44.68	54.46	$48.69_{\mathbf{+1.20}}$
SMA	A-T	Const	30.65	36.79	57.37	50.78	$43.90_{\mathbf{+2.29}}$	43.60	70.89	44.73	54.74	$48.69_{\mathbf{+1.20}}$

uniform data may not reduce noise as effectively as a near-zero learning rate, which could explain its slightly lower performance reported in Table 1. However, when using a curriculum strategy, labeled as *Ascend+EMA*, although the training starts with higher noise, the high-quality data introduced late in training provides a clearer and more reliable gradient. This strategy maintains the update magnitude in the high-quality regime, allowing it to take advantage of the good signal near the end along the signal direction. Using model averaging can also reduce noise along the noise direction, probably achieving a better balance between progress and stability compared to aggressive learning rate decay. We also provide a simplified theoretical model in Section 6 and discuss our perspective in the context of prior work (Wen et al., 2025) in Section B.1.

5.3 Results on Mid-Training with Mixed Quality Data

CMA Benefits are More Pronounced in Mid-Training. Mid-training is an emerging practice in LLM pretraining where a large corpus of average-quality data is supplemented by a smaller, high-quality dataset in a later training stage (Yang et al., 2025; OLMo et al., 2025; Hu et al., 2024). We conduct mid-training experiments, with settings detailed in Section A.1. As shown in Table 2, CMA exhibits a larger benefit in this practical setting compared to experiments on uniformly high-quality data. The CMA results (e.g., *EMA + A-T*) show a significant advantage over the WSD schedule baseline (*WSD + U,U*), improving the average accuracy by 1.20% and achieving over a 2.0% improvement on average across the core suite of benchmarks. This margin is notable given that no additional complex data filtering is applied. A possible explanation for the more prominent improvement is that, when high-quality data is sparse, each sample provides a relatively more valuable signal for parameter updates, amplifying the benefit of CMA.

A Practical and Simplified Strategy also Performs Well. In practice, sorting an entire data corpus globally may not be feasible. As an alternative, we tested a strategy where data is sorted in ascending order within each training phase separately (*A,A* in Table 2). The results show that the benefits of our approach over standard LR decay largely persist. However, applying a curriculum only to the final, high-quality data phase (*EMA + U,A*) is not sufficient for optimal results. The superior performance of the *A,A* schedule over *U,A* suggests that applying a curriculum to the initial, lower-quality data phase is also beneficial. One possible explanation is that reordering data in the lower-quality phase can exploit the model's forgetting mechanism to mitigate the adverse effects of toxic samples that pass through the cleaning pipeline by chance. These results further confirm the synergy between model averaging and a data curriculum.

5.4 Overlooked Benefit: Co-Design of Data Curriculum, LR Schedule, and Weight Average

CDMA: Combining Moderate LR Decay with Model Averaging. We have identified the benefits of the moderate LR decay and weight averaging in curriculum-based pretraining. A natural question is whether combining a moderate LR decay with model averaging under a data curriculum can yield further improvements. We conducted a series of experiments varying the ending learning rate in WSD schedules, from 1×10^{-5} to 3×10^{-3}, and then applied EMA to the final checkpoints. As shown in Figure 5, this combination achieves stable and optimal results with a moderate LR de-